KCBS Contextuality Under Non-Depolarizing Noise: Analytic Thresholds for Dephasing and Amplitude Damping

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(Dated: October 16, 2026)

We derive closed-form expressions for the Klyachko–Can–Binicioğlu–Shumovsky (KCBS) witness $\mathcal{K}(p)$ under two noise channels that have not previously been analyzed analytically for a spin-1 qutrit: dephasing and full-cascade amplitude damping. For the KCBS-optimal state $|2\rangle\langle 2|$, dephasing leaves the witness invariant ($\mathcal{K}(p) = \sqrt{5}$ for all p) because the optimal state carries no coherences. Amplitude damping yields $\mathcal{K}(p) = A\,p(2-p) + \sqrt{5}\,(1-p)^2$, where $A = \frac{5}{2} - \frac{\sqrt{5}}{2}$, with noise threshold $p^*_{\mathrm{AD}} = 1 - \sqrt{(\sqrt{5}-1)/(3\sqrt{5}-5)} \approx 0.149$. This is 2.78× lower than the depolarizing threshold $p^*_{\mathrm{dep}} \approx 0.415$, showing that the geometry of the noise channel — not only its strength — determines how quickly contextuality is lost. All results are verified numerically to machine precision.

INTRODUCTION

Quantum contextuality — the impossibility of assigning predetermined outcomes to measurements independently of which compatible measurements are jointly performed — is a foundational non-classical feature of quantum mechanics [1, 2]. The Klyachko–Can–Binicioğlu–Shumovsky (KCBS) inequality provides a particularly clean operational test for a spin-1 (qutrit) system: five sequential measurements whose correlations satisfy $\mathcal{K} \leq 2$ for any noncontextual model, while quantum mechanics predicts $\mathcal{K} = \sqrt{5} \approx 2.236$ for an optimal pure state [3].

Real experiments are subject to noise, and a natural question is: how much noise can the system tolerate before the quantum violation disappears? For depolarizing noise, the answer is known analytically [3, 4]: the KCBS witness decreases linearly with the noise parameter and the violation survives up to $p^*_{\mathrm{dep}} \approx 0.415$. Shaham and Eisenberg tested dephasing and two other unital channels experimentally on a biphoton qutrit, finding that dephasing suppresses the violation at a noise level of approximately $P \sim 0.17$ (read from a graph; no analytic formula was given) [5]. Dutra, Baldijão, and Terra Cunha recently extended the depolarizing analysis to sequential measurement noise in both KCBS and Peres–Mermin scenarios, and explicitly named dephasing as a physically relevant alternative channel without analyzing it [4].

In this paper we fill that gap. We derive analytic expressions for $\mathcal{K}(p)$ under dephasing and full-cascade amplitude damping on a single spin-1 qutrit, identify the noise thresholds in closed form, and compare all three channels. The results show that the structure of the noise — specifically, whether it acts on coherences, populations, or both — determines the robustness of contextuality in a geometrically transparent way.

SETUP

KCBS scenario

The system is a qutrit with Hilbert space $\mathcal{H} = \mathbb{C}^3$. Following Klyachko et al. [3], we define five rank-one projectors $P_i = |v_i\rangle\langle v_i|$, $i = 0, \ldots, 4$, whose unit vectors lie on a cone:

$$|v_i\rangle = \sin\theta\cos\phi_i\,|0\rangle + \sin\theta\sin\phi_i\,|1\rangle + \cos\theta\,|2\rangle, \quad (1)$$

where $\cos^2\theta = 1/\sqrt{5}$ and $\phi_i = 4\pi i/5$. Adjacent projectors satisfy $P_i P_{i+1 \bmod 5} = 0$.

The KCBS witness is

$$\mathcal{K}(\rho) = \sum_{i=0}^{4} \mathrm{Tr}(\rho\, P_i). \quad (2)$$

Any noncontextual hidden-variable model satisfies $\mathcal{K} \leq 2$. The optimal quantum state is $\rho_0 = |2\rangle\langle 2|$, which gives $\mathcal{K}(\rho_0) = \sqrt{5}$.

Noise model

We model noise as a completely positive trace-preserving (CPTP) map acting on the state before measurement. This is physically distinct from noise acting between sequential measurements; we discuss this distinction in Sec. . Three channels are compared:

Depolarizing.

$$\mathcal{D}_p(\rho) = (1-p)\,\rho + p\,\frac{\mathbb{I}}{3}, \quad p \in [0,1]. \quad (3)$$

Dephasing.

$$\mathcal{Z}_p(\rho)_{ij} = \begin{cases} \rho_{ij} & i = j, \\ (1-p)\,\rho_{ij} & i \neq j. \end{cases} \quad (4)$$

This is a unital channel that preserves populations and suppresses coherences.

Full-cascade amplitude damping. Models energy decay $|2\rangle \to |1\rangle \to |0\rangle$ with equal decay probability p per level. The Kraus operators are

$$\begin{aligned} K_0 &= \mathrm{diag}(1,\, \sqrt{1-p},\, (1-p)), \\ K_1 &= \sqrt{p}\,|0\rangle\langle 1|, \\ K_2 &= \sqrt{2p(1-p)}\,|1\rangle\langle 2|, \\ K_3 &= p\,|0\rangle\langle 2|. \end{aligned} \tag{5}$$

Completeness $\sum_k K_k^\dagger K_k = \mathbb{I}$ follows from the binomial identity $(1-p)^2 + 2p(1-p) + p^2 = 1$.

RESULTS

Depolarizing channel (baseline)

Substituting Eq. (3) into Eq. (2) and using $\mathrm{Tr}(\mathbb{I}/3 \cdot P_i) = 1/3$ for any rank-one projector gives

$$\mathcal{K}_{\mathrm{dep}}(p) = (1-p)\sqrt{5} + p\,\frac{5}{3}. \tag{6}$$

Setting $\mathcal{K}_{\mathrm{dep}}(p^*) = 2$ yields

$$p^*_{\mathrm{dep}} = \frac{\sqrt{5}-2}{\sqrt{5}-5/3} \approx 0.4146. \tag{7}$$

This reproduces the known result [3, 4].

Dephasing channel

Lemma 1. *$\mathcal{K}_{\mathrm{deph}}(p) = \sqrt{5}$ for all $p \in [0,1]$. The KCBS violation is fully robust to dephasing on the optimal state.*

Proof. The optimal state $\rho_0 = |2\rangle\langle 2|$ is diagonal in the standard basis: $(\rho_0)_{ij} = \delta_{i2}\delta_{j2}$. All off-diagonal elements are zero, so $\mathcal{Z}_p(\rho_0) = \rho_0$ for all p. Therefore $\mathcal{K}(\mathcal{Z}_p(\rho_0)) = \mathcal{K}(\rho_0) = \sqrt{5}$. □

This result is specific to the KCBS-optimal state. A state with nonzero coherences would be affected by dephasing; the immunity here reflects the geometric fact that $|2\rangle$ is the north pole of the projector cone defined in Eq. (1).

Full-cascade amplitude damping

Applying the channel (5) to $\rho_0 = |2\rangle\langle 2|$ gives a diagonal output state (amplitude damping preserves diagonality when the input is diagonal):

$$\mathcal{A}_p(\rho_0) = p^2|0\rangle\langle 0| + 2p(1-p)|1\rangle\langle 1| + (1-p)^2|2\rangle\langle 2|. \tag{8}$$

To evaluate the witness, we need $\sum_i \left|v_i^{(k)}\right|^2$ for each component $k \in \{0,1,2\}$ of the projector vectors. From Eq. (1):

$$\sum_{i=0}^{4} \left|v_i^{(0)}\right|^2 = \sum_{i=0}^{4} \left|v_i^{(1)}\right|^2 = \sin^2\theta \cdot \frac{5}{2} = \frac{5}{2} - \frac{\sqrt{5}}{2} \equiv A, \tag{9}$$

$$\sum_{i=0}^{4} \left|v_i^{(2)}\right|^2 = 5\cos^2\theta = \sqrt{5}. \tag{10}$$

The azimuthal sums $\sum_i \cos^2\phi_i = \sum_i \sin^2\phi_i = 5/2$ follow from the five-fold symmetry of the $\phi_i = 4\pi i/5$ angles.

Substituting Eq. (8) into Eq. (2):

$$\begin{aligned} \mathcal{K}_{\mathrm{AD}}(p) &= p^2 \cdot A + 2p(1-p) \cdot A + (1-p)^2 \cdot \sqrt{5} \\ &= A\,p(2-p) + \sqrt{5}\,(1-p)^2, \end{aligned} \tag{11}$$

where $A = 5/2 - \sqrt{5}/2$.

Setting $\mathcal{K}_{\mathrm{AD}}(p^*) = 2$ gives the quadratic $(B-A)\,p^2 - 2(B-A)\,p + (B-2) = 0$ with $B = \sqrt{5}$, $C \equiv B - A = 3\sqrt{5}/2 - 5/2$. The physical root (in $[0,1]$) is

$$p^*_{\mathrm{AD}} = 1 - \sqrt{\frac{\sqrt{5}-1}{3\sqrt{5}-5}} \approx 0.1493. \tag{12}$$

COMPARISON AND DISCUSSION

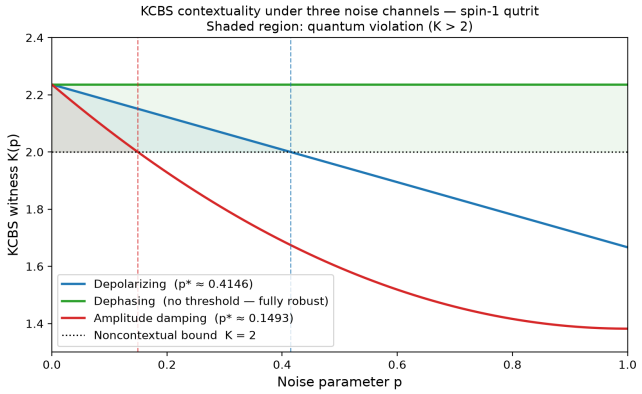


FIG. 1. KCBS witness $\mathcal{K}(p)$ under three noise channels on the optimal qutrit state $|2\rangle\langle 2|$. Shaded regions indicate quantum violation ($\mathcal{K} > 2$). Dashed vertical lines mark the noise thresholds for depolarizing ($p^* \approx 0.415$) and amplitude damping ($p^* \approx 0.149$).

The three results are summarized in Table I and Fig. 1.

The ratio $p^*_{\mathrm{dep}}/p^*_{\mathrm{AD}} \approx 2.78$ shows that amplitude damping destroys contextuality nearly three times faster than depolarization at equal noise parameter values. The physical reason is transparent from the geometry of the

Channel	$\mathcal{K}(p)$	p^*
Depolarizing	$(1-p)\sqrt{5} + p \cdot \frac{5}{3}$	0.4146
Dephasing	$\sqrt{5}$ (constant)	none
Amplitude damping	$A\,p(2-p) + \sqrt{5}(1-p)^2$	0.1493

TABLE I. KCBS witness and noise threshold for each channel on $\rho_0 = |2\rangle\langle 2|$. $A = 5/2 - \sqrt{5}/2 \approx 1.382$.

KCBS scenario. The witness $\mathcal{K}$ is maximized by $|2\rangle\langle 2|$, which sits at the north pole of the projector cone. Depolarizing noise moves the state uniformly toward the maximally mixed state $\mathbb{I}/3$, which still has $\mathcal{K}(\mathbb{I}/3) = 5/3 > 0$ — a relatively mild degradation. Amplitude damping funnels all population toward the ground state $|0\rangle\langle 0|$, which is the antipodal point on the cone and gives $\mathcal{K}(|0\rangle\langle 0|) = A \approx 1.382$ — well below the noncontextual bound. The state therefore crosses the bound much sooner.

Dephasing is inert on ρ_0 for a different reason: it acts only on off-diagonal elements, and ρ_0 has none. This is not a general statement about dephasing and KCBS contextuality — a superposition state would be affected — but it is a precise statement about the optimal state, and it has a direct experimental implication: an experiment that prepares $|2\rangle$ and applies dephasing noise will not see any degradation of the KCBS witness due to that noise alone.

Noise model caveat. All results here model noise as acting on the quantum state before measurement (Schrödinger picture, state noise). This is physically distinct from noise acting between sequential measurements, which was analyzed for the depolarizing channel in [4]. The two situations can give different thresholds and represent different experimental scenarios; the distinction must be stated explicitly in any experimental implementation.

Comparison with Shaham and Eisenberg [5]. Their experiment found that dephasing suppresses KCBS contextuality at $P \sim 0.17$ on a biphoton qutrit. Our result — that dephasing has no effect on the spin-1 optimal state — is not in contradiction. The biphoton system is physically different: noise acts independently on each photon, not on the qutrit as a whole, and the KCBS projectors are constructed differently. The two results together show that the robustness of contextuality to dephasing depends sensitively on the physical realization of the qutrit.

CONCLUSION

We have derived analytic expressions for the KCBS witness under dephasing and full-cascade amplitude damping on a spin-1 qutrit. The dephasing result is a lemma: the KCBS-optimal state is immune to dephasing because it carries no coherences. The amplitude damping result is the main contribution: a closed-form $\mathcal{K}(p)$ and threshold $p^*_{\mathrm{AD}} \approx 0.149$, derived from the binomial cascade structure of the Kraus operators and the five-fold symmetry of the KCBS projectors.

The comparison across three channels shows that the geometry of the noise — which degrees of freedom it acts on — matters as much as its strength. Amplitude damping is nearly three times more destructive to KCBS contextuality than depolarization at equal noise parameter, while dephasing is completely harmless on the optimal state.

These results provide analytic benchmarks for experimental implementations of KCBS tests in systems where amplitude damping is the dominant noise process, such as nitrogen-vacancy centers, trapped ions, and superconducting transmons operating as effective qutrits.

Computations were performed using NumPy [6]. Code and data are available at `https://github.com/mariohgh/kcbs-noise-thresholds`.

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